

RIYADH KING SALMAN AB, Saudi Arabia

WMO: 404380

Lat: 24.722N

Lon: 46.725E

Elev: 2084

StdP: 13.62

Time Zone: 3.00 (E03)

Period: 94-14

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	43.2	46.3	9.4	9.6	63.3	13.7	11.9	69.5	21.2	62.4	19.6	63.5	4.6	320	0.477

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	23.5	112.8	67.1	111.2	66.3	109.7	65.7	70.6	99.8	69.1	100.2	67.9	100.7	9.8	30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.3	97.6	73.4	62.2	90.4	71.8	59.3	81.4	72.0	35.7	100.9	34.4	99.2	33.4	100.4	80.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.7	18.3	16.2	DB	38.0	115.3	3.7	1.5	35.4	116.4	33.2	117.3	31.1	118.2	28.4	119.3	(4)
(5)			WB	31.2	74.7	3.3	2.7	28.9	76.7	26.9	78.3	25.1	79.8	22.7	81.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	80.7	59.0	64.1	71.0	81.2	91.3	96.3	98.0	98.2	92.3	83.1	70.7	61.7		
	DBStd	14.99	5.71	6.48	6.35	6.13	4.51	2.77	2.77	2.71	3.45	4.09	5.82	6.01		
	HDD50	8	5	1	0	0	0	0	0	0	0	0	0	2		
	HDD65	457	201	86	17	1	0	0	0	0	0	0	18	134		
	CDD50	11199	283	397	652	936	1282	1389	1488	1493	1268	1025	620	366		
	CDD65	6173	14	62	204	487	817	939	1023	1028	818	560	187	34		
	CDH74	99410	129	568	2040	6261	13417	16721	18487	18572	13731	7566	1657	261		
	CDH80	68527	16	152	832	3420	9167	12405	14023	14109	9525	4274	571	33		
(14) Wind	WSAvg	7.2	7.0	8.1	8.1	8.5	7.7	8.1	7.9	7.1	6.1	5.4	6.3	6.5		
(15) Precipitation	PrecAvg	4.9	0.6	0.3	1.2	1.5	0.3	0.0	0.0	0.0	0.0	0.0	0.3	0.4		
	PrecMax	12.5	4.4	1.5	4.8	5.2	2.7	0.0	0.4	0.2	0.0	1.1	3.7	2.8		
	PrecMin	0.6	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
	PrecStd	3.5	0.9	0.4	1.4	1.6	0.6	0.0	0.1	0.0	0.0	0.2	0.7	0.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	82.3	89.2	96.0	104.2	110.9	113.4	115.0	114.5	109.9	102.2	92.5	83.8		
		MCWB	57.8	61.1	60.7	63.8	65.2	66.3	67.8	68.3	66.2	63.2	62.0	59.7		
	2%	DB	77.4	84.5	91.8	100.5	107.9	111.3	112.9	112.2	108.0	99.4	89.4	80.4		
		MCWB	55.9	58.3	60.1	63.3	64.9	65.7	66.9	67.5	65.6	62.1	60.3	58.4		
	5%	DB	73.7	80.7	88.2	98.2	105.9	109.6	111.2	111.1	106.1	97.2	86.3	76.9		
		MCWB	54.0	57.0	58.8	62.5	64.7	65.1	66.0	66.8	64.9	61.5	59.5	56.7		
	10%	DB	70.1	77.1	84.5	94.7	103.7	107.8	109.4	109.5	104.2	95.2	83.3	73.5		
		MCWB	53.2	55.0	57.4	61.9	64.4	65.5	66.3	66.3	64.1	61.3	58.8	55.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	63.1	63.9	67.4	69.8	70.8	69.1	72.1	74.2	69.9	68.3	67.8	65.6		
		MCDB	71.0	76.1	79.4	83.3	90.4	107.2	103.0	98.9	96.1	81.7	77.2	73.9		
	2%	WB	60.2	61.7	64.1	67.6	68.5	67.3	69.5	70.9	67.6	65.9	65.3	62.8		
		MCDB	69.0	75.3	79.3	84.7	95.0	107.6	106.8	104.7	102.3	86.5	75.0	70.1		
	5%	WB	58.1	59.6	62.1	66.0	67.2	66.2	67.7	69.2	66.3	64.2	63.4	60.6		
		MCDB	68.3	75.1	78.2	84.9	96.6	107.0	107.6	105.1	102.3	89.8	75.4	71.1		
	10%	WB	55.7	57.5	60.2	64.5	65.8	65.1	66.4	67.9	65.2	62.9	61.8	58.2		
		MCDB	66.2	73.8	77.2	86.0	97.1	105.9	107.5	105.8	101.4	90.4	76.3	69.4		
(35) Mean Daily Temperature Range	MDBR		18.9	20.4	21.4	21.8	22.5	23.6	23.1	23.5	23.7	23.3	19.9	19.2		
	5% DB	MCDBR	24.1	24.7	26.2	25.4	24.5	25.1	24.0	24.0	24.6	25.0	24.0	23.9		
		MCWBR	11.6	10.9	11.0	9.1	8.3	8.5	8.3	8.2	8.6	9.3	9.9	11.0		
	5% WB	MCDBR	18.5	21.0	20.6	21.4	21.5	23.4	22.5	22.1	22.9	21.4	17.7	17.3		
		MCWBR	10.7	10.4	10.7	8.6	7.6	8.3	8.4	8.3	8.4	9.1	8.0	9.1		
(40) Clear-Sky Solar Irradiance	taub		0.427	0.471	0.541	0.622	0.644	0.603	0.583	0.574	0.517	0.450	0.461	0.423		
	taud		1.980	1.854	1.697	1.572	1.531	1.586	1.655	1.675	1.771	1.923	1.904	1.990		
	Ebn at Noon		257	254	243	227	222	230	234	236	246	257	243	251		
	Edh at Noon		51	62	76	88	91	86	80	78	69	57	54	49		
(44) All-Sky Solar Radiation	RadAvg		1366	1638	1880	2046	2281	2485	2373	2260	2078	1812	1403	1295		
	RadStd		96	93	103	101	90	76	54	50	47	68	98	86		

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	+1.57	N/A	N/A	N/A	+450	+468
(47)	Regional (1 neighbor)	+0.59	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+202	+185

Nomenclature: See separate page